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PROJECT REPORT

ON

Mobile App for Direct Market Access for Farmers

Project -I (BTCS-603-18)

BACHELOR OF TECHNOLOGY
(Robotic & AI)

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Mobile App for Direct Market Access for Farmers

Abstract

Agriculture plays a crucial role in the economy, yet many farmers face difficulties in receiving fair prices for their produce due to the involvement of multiple intermediaries in the supply chain. This project proposes the development of a **mobile application that provides direct market access for farmers**, allowing them to connect directly with buyers such as retailers, wholesalers, and consumers.

The application will enable farmers to list their products, update quantities, and set prices, while buyers can browse available products and place orders directly through the platform. This approach reduces dependency on middlemen and ensures better price transparency and profit for farmers.

The application will be developed using **React Native for mobile development**, **Node.js for backend services**, and **SQL for database management**. Additional features such as **price prediction, demand analysis, and smart product recommendations using basic AI techniques** will be incorporated to assist farmers in making informed selling decisions.
The proposed system aims to improve agricultural marketing efficiency, enhance farmers' income, and promote digital transformation in the agricultural sector.

Introduction

Agriculture is one of the most important sectors of the economy in many countries, especially in developing nations. A large portion of the population depends on agriculture as their primary source of income and employment. Farmers play a crucial role in ensuring food security by producing crops that feed millions of people. Agricultural production supports not only the food supply but also contributes significantly to national economic growth, employment generation, and rural development.

Despite their importance, farmers often face numerous challenges that prevent them from earning fair profits from their hard work. Many farmers struggle with unstable market prices, lack of proper infrastructure, limited access to modern technologies, and insufficient market information. As a result, farmers frequently sell their produce at lower prices and are unable to fully benefit from the agricultural value chain.

One of the major issues in agricultural marketing is the presence of multiple intermediaries between farmers and consumers. These intermediaries include commission agents, traders, wholesalers, and retailers. While they play a role in the distribution process, they also take a significant share of the profit. Because of this long supply chain, the price paid by the final consumer is often much higher than the amount received by the farmer.

Farmers also face difficulties in accessing reliable market information. Many farmers do not know the current market demand, price trends, or potential buyers in different regions. Without this information, they are forced to depend on local traders who may offer lower prices for their crops. In addition, farmers may lack proper storage facilities and transportation systems, which forces them to sell their produce quickly after harvesting, even if the prices are not favorable.

In recent years, technological advancements have created new opportunities to improve agricultural systems. The widespread use of smartphones and mobile internet has opened new possibilities for digital platforms that can connect farmers directly with buyers. Mobile applications have the potential to transform agricultural marketing by reducing dependency on intermediaries and providing farmers with better access to markets.

The proposed **Mobile App for Direct Market Access for Farmers** aims to develop a digital platform that allows farmers to directly interact with buyers such as retailers, wholesalers, and consumers. Through this platform, farmers can list their products, manage their inventory, and negotiate prices with buyers. Buyers can browse available products, compare prices, and place orders directly from their mobile devices.

In addition to providing a marketplace, the system will integrate artificial intelligence (AI) to assist farmers in making better decisions. The AI component can analyze historical market data, predict price trends, and suggest the best time to sell specific crops. This can help farmers maximize their profits and reduce financial uncertainty.

Overall, the proposed mobile application aims to modernize the agricultural supply chain, increase transparency in the market, and improve the economic condition of farmers by giving them direct access to buyers.

Agricultural marketing refers to the process through which agricultural products move from farms to consumers. This process includes harvesting, storage, transportation, distribution, and

selling of agricultural goods. In traditional agricultural marketing systems, several intermediaries are involved in the supply chain.

The typical supply chain often includes the following participants:

- Farmers
- Local traders
- Commission agents
- Wholesalers
- Retailers
- Consumers

Each participant in the supply chain adds their own margin, which increases the final price of the product. However, the farmer usually receives only a small portion of the total price paid by the consumer.

Another problem in traditional systems is the lack of transparency. Farmers often rely on middlemen to determine the price of their produce. These intermediaries may not always provide accurate information about market conditions, which can lead to unfair pricing.

Furthermore, rural farmers often face limited access to transportation facilities. Due to poor infrastructure and high transportation costs, farmers prefer to sell their produce locally even if prices are lower. This reduces their overall income.

The traditional agricultural system therefore creates multiple barriers that prevent farmers from accessing larger markets and earning better profits.

Challenges Faced by Farmers

Farmers encounter several challenges in the agricultural marketing system. Some of the major challenges include:

Lack of Market Information

Many farmers are not aware of the latest market prices of their crops in different markets. Without proper information, farmers cannot make informed decisions about when and where to sell their produce.

Price Fluctuations

Agricultural prices often fluctuate due to supply and demand conditions. Farmers may experience sudden drops in prices during harvest seasons when supply is high.

Storage Limitations

Many farmers do not have access to proper storage facilities. As a result, they must sell their crops immediately after harvesting, even if the market price is low.

Transportation Issues

Transportation is another major problem for farmers, particularly in rural areas. Poor road infrastructure and high transportation costs make it difficult for farmers to access distant markets.

Dependence on Intermediaries

Farmers often rely on intermediaries to sell their products. These intermediaries take a portion of the profit, which reduces the income of farmers.

Role of Mobile Technology in Agriculture

Mobile technology has become widely accessible in rural areas due to the increasing availability of affordable smartphones and internet connectivity. Mobile applications can provide farmers with valuable information and services that improve agricultural productivity and marketing.

Some of the key benefits of mobile technology in agriculture include:

- Access to real-time market prices
- Weather forecasts and agricultural advice
- Direct communication with buyers
- Online product listing and order management

Mobile apps provide a convenient and user-friendly interface that farmers can easily use without requiring advanced technical knowledge.

Literature survey

In recent years, several researchers, government organizations, and private companies have attempted to improve the agricultural marketing system through the use of digital technologies. The goal of these efforts is to reduce inefficiencies in the supply chain, improve transparency in pricing, and provide farmers with better access to markets. Many digital platforms and research studies have explored different approaches to connect farmers with buyers, provide market information, and improve agricultural productivity.

A review of existing systems and research work helps in understanding the current technological solutions available in the agricultural sector and also highlights the limitations that still exist. This literature survey discusses some of the major platforms, applications, and research studies related to digital agricultural marketplaces and farmer support systems.

Electronic National Agriculture Market (eNAM)

One of the most significant initiatives in India aimed at improving agricultural marketing is the **Electronic National Agriculture Market (eNAM)** platform. Enam is an online trading portal developed by the Government of India to connect multiple Agricultural Produce Market Committees (APMCs) across the country. The main objective of this platform is to create a unified national market for agricultural commodities.

Through the eNAM system, farmers can list their produce online and participate in digital auctions conducted by traders. Buyers from different regions can view the available agricultural products and place bids through the online platform. This system helps improve price discovery and allows farmers to receive competitive prices for their produce.

The eNAM platform also improves transparency in the trading process. Farmers are able to view real-time market prices and trading activities, which helps them make better decisions about selling their crops. Additionally, the system reduces the need for physical presence in the market yard and speeds up the trading process.

However, despite these advantages, the eNAM platform still operates within the framework of regulated agricultural markets. Farmers are usually required to bring their produce to registered market yards, and transactions often involve licensed traders and commission agents. Therefore, the platform does not completely eliminate intermediaries in the agricultural supply chain. This limitation highlights the need for systems that allow farmers to sell their produce directly to buyers without relying on traditional market structures.

AgriMarket Information System

Another important digital initiative designed to support farmers is the **AgriMarket application**. This system primarily focuses on providing farmers with updated information about agricultural commodity prices in various markets.

The AgriMarket platform collects market price data from different agricultural markets and presents it in an easy-to-understand format. Farmers can use this information to compare prices in nearby markets and identify where they can sell their produce at better rates. Access to accurate price information helps farmers avoid exploitation by middlemen who may offer lower prices due to information asymmetry.

The application also helps farmers monitor price trends over time. By observing price fluctuations, farmers can decide the most suitable time to sell their crops. This can significantly improve their chances of receiving fair prices for their produce.

While the AgriMarket system is useful in providing price information, it has certain limitations. The platform mainly serves as an information provider and does not offer direct trading capabilities. Farmers cannot directly connect with buyers or complete transactions through the application. As a result, farmers still need to rely on traditional marketing channels to sell their crops.

Therefore, although AgriMarket improves price awareness among farmers, it does not fully solve the problem of direct market access.

Gaps in existing projects

The literature survey shows that many digital platforms and applications have been developed to support farmers and improve the agricultural marketing system. These platforms aim to provide market information, agricultural inputs, and various advisory services to farmers. Government initiatives such as digital agricultural markets and private agri-tech platforms have contributed significantly to modernizing agricultural practices.

However, despite these advancements, several limitations and gaps still exist in the current systems. These gaps reduce the overall effectiveness of digital platforms in solving the real problems faced by farmers in agricultural marketing. Identifying these gaps is important in order to design a new system that can provide better solutions and improve farmers' access to markets.

The following sections describe the major gaps identified in the existing agricultural digital platforms.

Lack of Direct Selling Mechanism

Another important limitation observed in many current agricultural applications is the absence of a direct selling mechanism. Some platforms provide useful services such as market price information, weather forecasts, or farming advice. While these services help farmers make better decisions regarding crop cultivation, they do not address the problem of selling agricultural products directly.

For example, certain applications allow farmers to view price information from different markets. Although this helps farmers understand market trends, they still need to physically visit markets or depend on traders to sell their produce. The absence of an integrated buying and selling system means that farmers cannot complete transactions through the platform itself.
A true digital marketplace should allow farmers to upload product details, display their available crops, and interact directly with buyers who are interested in purchasing those products. The lack of such functionality in many existing platforms limits their effectiveness in improving agricultural marketing.

Complex and Non-User-Friendly Interfaces

Many digital platforms are designed with complex interfaces that may be difficult for farmers to use. Farmers, particularly those in rural areas, may have limited experience with advanced technology or digital applications.

If a mobile application includes too many complicated features or requires extensive technical knowledge, farmers may find it difficult to operate. This can discourage them from using the system regularly.

A successful agricultural mobile application should have a simple and intuitive interface that farmers can easily understand and use without requiring technical expertise. The design should focus on clear navigation, simple instructions, and minimal complexity so that farmers can quickly learn how to use the platform.

The lack of user-friendly interfaces in many existing systems remains a significant barrier to the widespread adoption of digital agricultural technologies.

Lack of Intelligent Decision Support Systems

Another major gap in existing agricultural platforms is the absence of intelligent decision support systems. Most platforms provide basic information such as market prices or weather conditions but do not offer personalized recommendations to farmers.

Farmers often face uncertainty when deciding the best time to sell their crops or determining the appropriate price for their products. Without proper guidance, farmers may sell their produce at lower prices or miss opportunities to earn higher profits.

Advanced technologies such as artificial intelligence and data analytics can be used to analyze historical market data and identify patterns in agricultural markets. By using these technologies, digital platforms can provide valuable recommendations to farmers.

For example, an intelligent system could suggest the optimal selling price for a crop based on current market conditions and past price trends. It could also provide predictions about future demand and supply conditions.

The absence of such intelligent features in many existing platforms limits their ability to support farmers in making informed decisions..

Need for an Improved Digital Solution

The analysis of existing platforms clearly shows that while digital technologies have improved certain aspects of agricultural services, significant gaps still remain in the agricultural marketing system.

To address these challenges, there is a need for a comprehensive digital platform that focuses specifically on improving direct market access for farmers. Such a system should eliminate unnecessary intermediaries and create a direct connection between farmers and buyers.

The proposed **Mobile App for Direct Market Access for Farmers** aims to address the identified gaps by incorporating the following features:

- Direct communication between farmers and buyers
- Easy product listing and ordering system
- AI-based price recommendation system
- Real-time demand analysis
- Simple and user-friendly mobile interface

Problem Formulation

Agriculture plays a vital role in supporting the economy and ensuring food security. A large number of farmers depend on agricultural activities as their primary source of income. Despite their hard work and contribution to food production, many farmers face serious difficulties in earning fair profits from their agricultural produce. One of the major reasons for this issue is the traditional agricultural marketing system, which involves multiple intermediaries between farmers and consumers.

In the current system, farmers usually sell their crops to local traders, commission agents, or wholesalers who then distribute the products to retailers and consumers. These intermediaries take a significant portion of the profit, which reduces the income that farmers receive. As a result, the price paid by consumers is often much higher than the price received by the farmers who produce the crops.

Another important problem faced by farmers is the lack of accurate and timely market information. Many farmers are not aware of current market prices, demand trends, or potential buyers in different markets. Due to this lack of information, farmers may sell their produce at lower prices or in less profitable markets. In addition, limited storage facilities and transportation challenges often force farmers to sell their crops immediately after harvesting, even when market prices are unfavorable.

Therefore, the main problem addressed in this project is the **absence of an efficient, simple, and accessible digital platform that enables farmers to directly connect with buyers and sell their agricultural products without intermediaries**. Without such a platform, farmers continue to face limited market access, reduced profits, and lack of transparency in the selling process.

The proposed solution to this problem is the development of a **Mobile App for Direct Market Access for Farmers**. This mobile application will provide a digital marketplace where farmers can list their agricultural products, specify quantities and prices, and interact directly with buyers such as retailers, wholesalers, and consumers. Buyers will be able to browse available products, compare prices, and place orders through the application.

By enabling direct communication and transactions between farmers and buyers, the proposed system aims to reduce the role of intermediaries, improve market transparency, and ensure that farmers receive fair prices for their produce. The platform will also simplify the selling process and provide farmers with better access to a wider market, ultimately helping to increase their income and improve the efficiency of the agricultural supply chain.

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OBJECTIVES

The main objectives of this project are:

1. To develop a **mobile application that connects farmers directly with buyers**.
2. To reduce the role of intermediaries in the agricultural supply chain.
3. To provide **real-time product listing and ordering system**.
4. To implement **AI-based price prediction and demand analysis**.
5. To create a **simple and user-friendly interface** for farmers.
6. To improve the profitability and efficiency of agricultural marketing.

Methodology and Planning

The development of the proposed **Mobile App for Direct Market Access for Farmers** will follow a structured and systematic software development process. The purpose of this methodology is to ensure that the system is developed efficiently and meets the requirements of both farmers and buyers. A well-defined development approach helps in identifying the necessary features, designing an effective system architecture, implementing the required functionalities, and testing the application before deployment.

The development process will consist of several important stages including requirement analysis, system design, development, artificial intelligence integration, testing, and deployment. Each stage plays a critical role in ensuring that the final system is reliable, efficient, and easy to use.

Requirement Analysis

The first stage in the development of the proposed system is requirement analysis. In this stage, the needs and expectations of the users are carefully identified and analyzed. The main users of the system will be farmers and buyers such as retailers, wholesalers, and consumers.

Farmers require a platform where they can easily list their agricultural products, specify quantities, set prices, and manage their inventory. The application should allow farmers to upload product information such as crop name, quantity available, quality details, and location. Farmers should also be able to update or remove product listings when necessary.

Requirement analysis helps ensure that all essential features are clearly defined before the development process begins. This stage forms the foundation for designing and implementing the system.

System Design

After identifying the system requirements, the next stage is system design. System design involves planning the overall structure and architecture of the application. This includes determining how different components of the system will interact with each other.

The proposed system will follow a three-layer architecture consisting of the frontend, backend, and database components.

The frontend is the part of the application that users interact with directly. It includes the user interface and all visual elements of the mobile application. The frontend will allow farmers and buyers to perform actions such as registering accounts, listing products, searching for items, and placing orders.

The backend is responsible for handling the core functionality of the application. It processes user requests, manages business logic, and communicates with the database. The backend ensures that data is securely stored and retrieved when needed.

The database is used to store important information such as user accounts, product listings, orders, and transaction records. The database must be structured efficiently so that data can be accessed quickly and reliably.

Frontend Development

The frontend of the mobile application will be developed using modern web and mobile technologies. The main technology used for mobile development will be **React Native**, which allows developers to create cross-platform mobile applications using JavaScript.

React Native provides a flexible and efficient framework for building mobile applications with responsive user interfaces. It enables developers to design applications that run smoothly on both Android and iOS devices.

Additional technologies such as **HTML**, **CSS**, and **JavaScript** will also be used to design and style the user interface. These technologies help create visually appealing layouts and interactive components that improve the user experience.

The frontend will include several key features such as user registration and login, product listing forms for farmers, product search functionality for buyers, order placement interfaces, and notification systems.

Backend Development

The backend of the application will be developed using **Node.js** and **Express.js**. Node.js is a powerful server-side runtime environment that allows developers to build scalable and high-performance applications. It uses JavaScript on the server side, which makes it compatible with frontend development technologies.

Express.js is a lightweight web framework that simplifies the process of building web applications and APIs. It helps manage routing, handle user requests, and process data efficiently.

The backend server will handle tasks such as user authentication, product management, order processing, and communication between the frontend and the database. It will also ensure that data is securely transmitted and stored.

Database Development

The application will use a **SQL-based database** to store all necessary information. The database will include several tables such as users, products, orders, and transactions.

The user table will store account details of farmers and buyers, including usernames, contact information, and login credentials. The product table will contain information about the agricultural products listed by farmers, such as crop name, quantity, price, and availability.

The orders table will record details of transactions between farmers and buyers, including order status and delivery information. Proper database design is essential to ensure efficient data management and fast system performance.

AI Integration

To enhance the functionality of the application, some basic artificial intelligence features will be integrated into the system. AI technologies can help farmers make better decisions regarding pricing, crop selection, and market trends.

Price Prediction

One of the AI features that will be implemented is price prediction. This feature will analyze historical market price data for various agricultural products. By studying past price trends, the system can estimate the optimal selling price for a particular crop.

Price prediction helps farmers determine competitive prices that attract buyers while still maximizing profits. It also helps farmers avoid selling their produce at significantly lower prices due to lack of market information.

Demand Prediction

Another important feature is demand prediction. This feature analyzes product search patterns and purchase trends within the application. By studying which products buyers are searching for most frequently, the system can identify crops that are currently in high demand.
This information can help farmers plan their future crop production and focus on crops that are likely to generate higher profits.

Smart Recommendation System

The system will also include a smart recommendation feature for buyers. This feature will analyze previous purchase behavior and suggest products that buyers may be interested in purchasing.

For example, if a buyer frequently purchases a specific type of crop, the system may recommend similar products from other farmers. This helps buyers find products quickly and increases the chances of successful transactions for farmers.

Deployment

After successful testing, the final stage of the development process is deployment. Deployment involves making the application available to users.
The mobile application will be deployed so that farmers and buyers can install it on their smartphones. The backend server will be hosted on a **cloud-based platform**, which ensures reliable performance and easy scalability.

Once deployed, the application will provide farmers with a digital marketplace where they can directly sell their agricultural products and connect with buyers without intermediaries. This will improve market access, increase transparency, and help farmers achieve better profits for their produce.

SOFTWARE REQUIREMENT SPECIFICATION

Hardware Requirements

- Smartphone with Android OS
- Internet connectivity
- Server for backend hosting

Software Requirements

Frontend:

- React Native
- HTML
- CSS
- JavaScript

Backend:

- Node.js
- Express.js

Database:

- MySQL / SQL

Tools:

- Visual Studio Code
- GitHub
- Postman

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